

MUHAMMAD TALAL

PROFESSIONAL SUMMARY

I'm Muhammad Talal, an AI student specializing in Machine learning, Computer vision, NLP, and LLM systems with hands-on experience building and deploying end-to-end ML solutions. Also experienced as a full-stack developer, building web and mobile applications using React, Node.js, Flutter, and React Native. Currently maintaining a 3.29 CGPA while actively solving real-world problems with AI.

PROFESSIONAL SKILLS

Programming

- Python, Java, C++
- Algorithms & Problem Solving
- Database Design (MySQL, MongoDB)

Artificial Intelligence

- Machine Learning
- Generative AI / LLMs (RAG, LangChain)
- Computer Vision
- Neural Networks (CNN, RNN, ANN)

Web & Development Tools

- HTML5, CSS3, JavaScript
- React.js, Node.js
- Flask, Django
- RESTful APIs,
- Flutter, React Native

EDUCATION

BS Artificial Intelligence | CGPA: 3.29/4.00

- NFC Institute of Engineering and Fertilizer Research, Faisalabad, Pakistan
- 2023- Ongoing

PROJECTS

Circuit Board Component Detection.

- Built a YOLO v5-based computer vision system to detect and classify 15+ electronic component types on circuit boards in real-time at 50 FPS with ~92% detection accuracy.
- Implemented end-to-end pipeline using PyTorch, OpenCV, and ONNX; trained on 1,000+ diverse board images with CSPDarknet backbone achieving 0.90 F1-score and edge deployment support.

BiEncoder Embedding Dimension Optimization.

- Conducted NLP research comparing 1024 vs 4096-dimensional embeddings in BiEncoder models for dense retrieval, evaluating across 2,800 queries on benchmark datasets.
- Demonstrated ~300% improvement in Recall@500 with 4096-dim embeddings using PyTorch and Hugging Face; delivered practical recommendations on dimension selection based on resource constraints.

Sentiment Classification Engine.

- Developed a Bidirectional LSTM model in TensorFlow/Keras to classify social media text as Positive, Neutral, or Negative, achieving ~85% test accuracy across 11 training epochs.
- Built a full preprocessing pipeline with NLTK covering noise removal, tokenization, stopword removal, and lemmatization; deployed a live interactive demo with real-time inference.

AI-Powered Disaster Management System

- Developed a cross-platform Disaster Management System comprising an Android mobile app and a PHP/MySQL web admin panel — enabling real-time disaster reporting, GPS-based help requests, emergency helpline access, and live disaster mapping with color-coded markers by disaster type.
- Integrated an AI-powered disaster risk prediction engine using a Random Forest Classifier trained on environmental features (rainfall, temperature, humidity, river level, earthquake magnitude) to classify risk levels with automated verification and notification pipelines.

AI Interview Conduct System.

- Built a real-time interview proctoring platform using Flask and OpenCV with 95% detection accuracy, featuring face detection, eye tracking, head orientation monitoring, and multi-person violation alerts at 30 FPS.
- Integrated Gemini 2.5 Flash API for AI-powered question generation from PDF input, speech-to-text transcription, intelligent answer evaluation, and AI-generated content detection (0–100 scoring).

Sales Intelligence & Business Analytics Dashboard

- Designed a 12-page Power BI business intelligence dashboard for Motorola sales data, tracking 769M in total revenue, 19K units sold, and 4K transactions with real-time KPI monitoring.
- Implemented multi-dimensional analytics using DAX, Power Query, and SQL Server covering product performance, payment method distribution (credit/debit card, cash), and temporal sales trends.

EXPERIENCE

Freelance AI Developer | Self-Employed

2025 – Present

- Developing and delivering full-stack web and mobile applications for clients, utilizing React, Node.js, Flutter, and React Native.
- Building AI-integrated solutions incorporating machine learning models, REST APIs, and database management (MySQL, MongoDB).
- Managing complete project lifecycles from requirements gathering and system design to deployment and client handover.
- Delivering cross-platform mobile apps and responsive web dashboards tailored to client business needs.

LANGUAGES

- English (Fluent)
- Urdu (Native)